SMART ENERGY MANAGEMENT USING FORECAST-DRIVEN GENERATION OPTIMIZATION.

INTRODUCTION

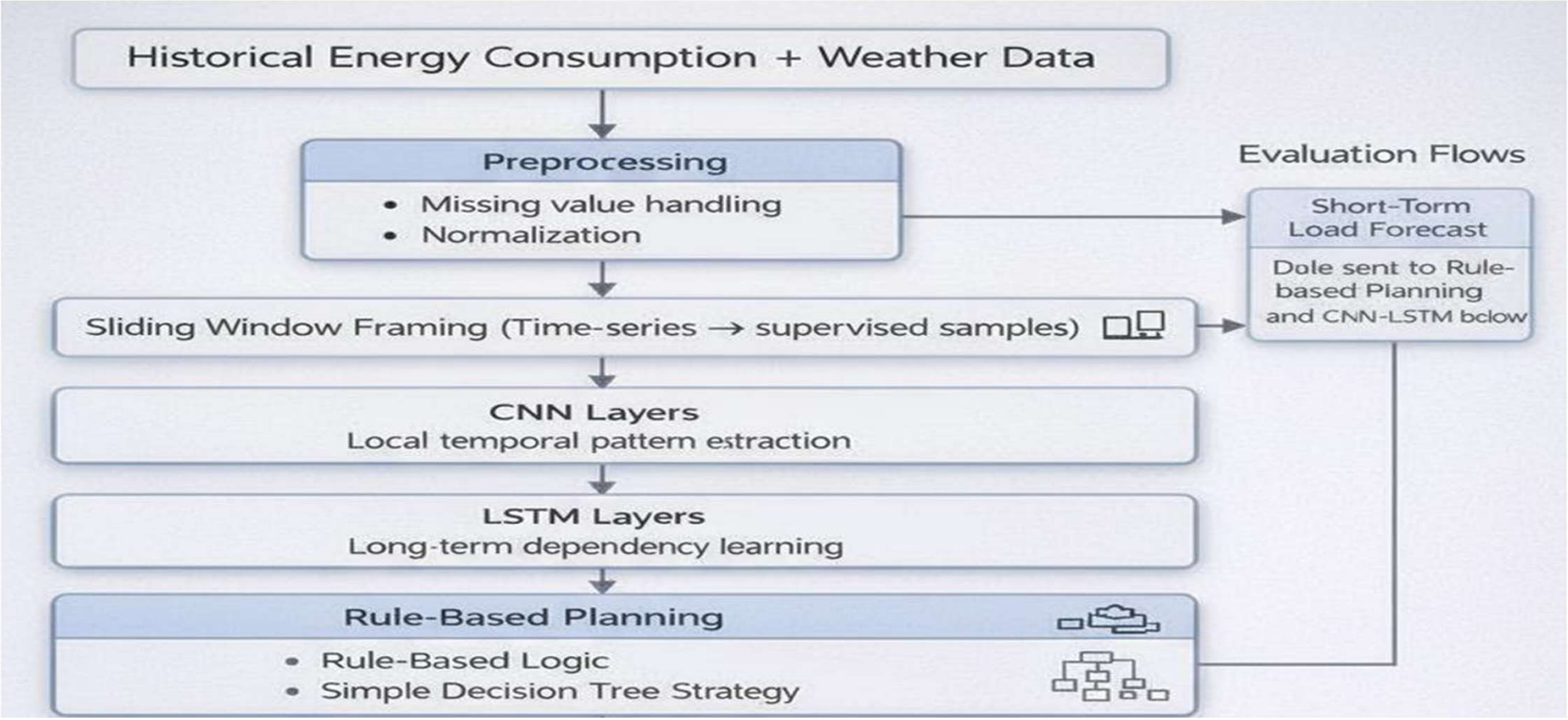
In modern power systems, electricity demand changes continuously due to factors such as **consumer usage patterns, weather conditions, and time of day**, while power generation must be scheduled in advance. Many traditional power systems still rely on **reactive control**, where generation adjustments are made only after demand increases, leading to **inefficient energy utilization, higher operating costs, and stress on grid infrastructure**.

To address this problem, our project develops a **machine learning–based forecasting framework** that predicts **electricity load, renewable generation, and electricity prices** using historical data, weather information, and time-based features from the **Open STEF Liander 2024 Energy Forecasting Benchmark dataset**, which contains **15-minute interval electricity load data from 55 grid locations along with weather and day-ahead electricity price data**.

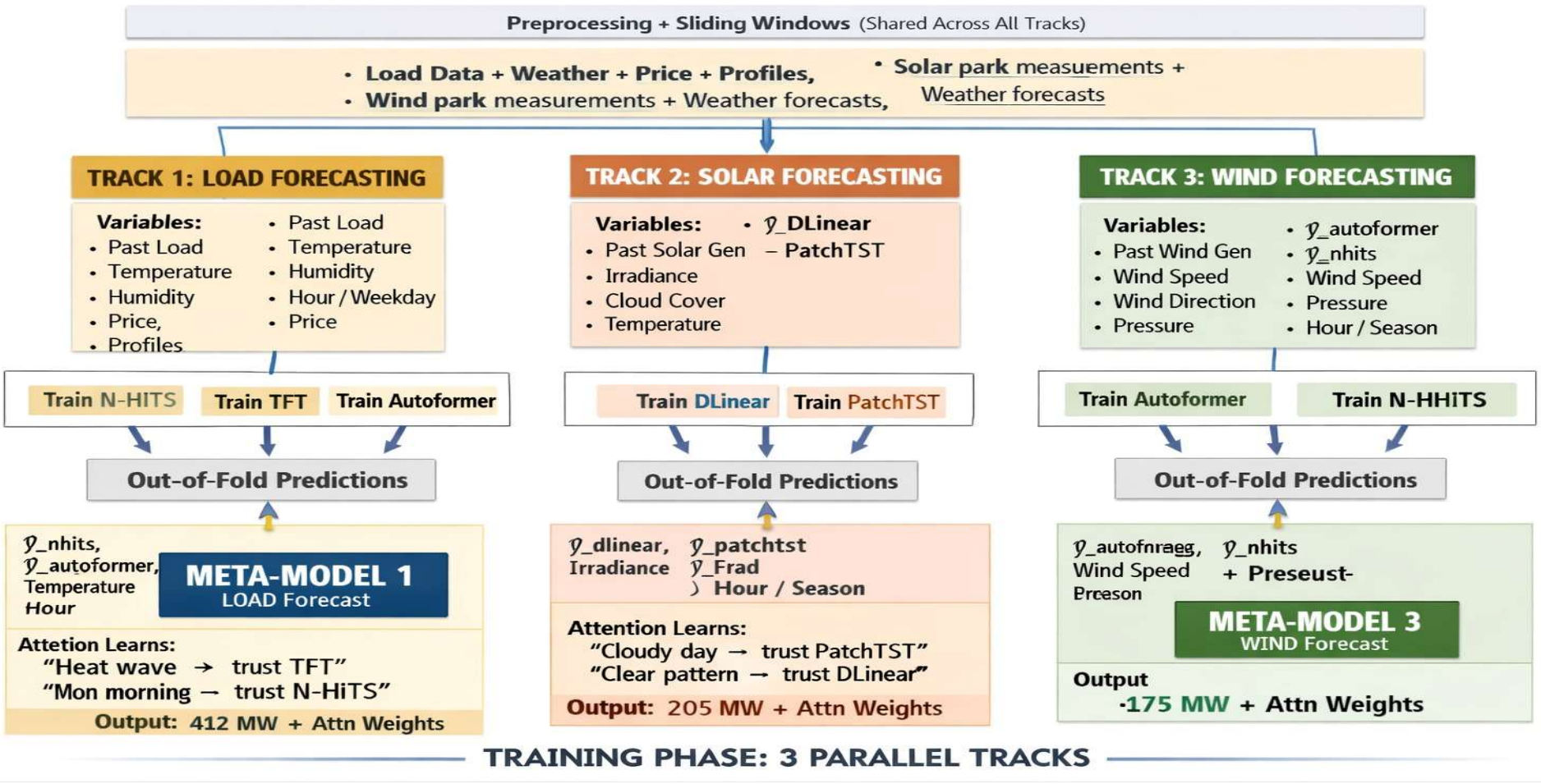
By forecasting both **energy demand and electricity cost**, the system estimates the **future net load and price risk**, enabling better **generation scheduling and efficient energy management**. The goal is to shift power systems from **reactive operation to predictive and cost-efficient energy planning**, improving **grid reliability and efficiency**.

MODEL ARCHITECTURE

OLD ARCHITECTURE DIAGRAM



MODEL ARCHITECTURE



THANK YOU